Claims Integrity and Risk Analysis

Summary insights

1. **There is evidence of potential irregularities in the billing and treatment patterns of seven patients that have billed the hospital for atypically high amounts.**
   Although these seven patients typically received basic or standard-level treatments, they each had a cost-per-appointment of at least $1,700 higher than the hospital-wide average. This suggests that the treatments given to the patients were abnormally high in cost. They also had multiple appointments with specialists in pediatrics, despite being aged over 16.

2. **More than half of the treatment costs billed to these 'high-risk' patients – a total of more than $80,000 – have not been paid, representing a significant liability for the hospital.**
   Four of these patients have failed billings of more than $7,600 on an individual level. This likely to be disproportionately affecting MedCare Plus, which is being billed for more than double the amount billed to any other insurance provider at the hospital ($1.4 million).

3. **Issues with data quality in the patient and appointment datasets mean that it is challenging to conclusively quantify the extent of risks in this area.**
   Problems identified include apparent duplication in address and email data, and potential irregularities in patient registration data.

Recommendation:

Target further investigations into the identified high-risk patients and the doctors responsible for treating them.

Two doctors (Linda Wilson, oncologist, and Sarah Taylor, dermatologist) are particularly likely to have treated the high-risk patients and were also among those billing the highest amounts to MedCare Plus. Further investigation could include scrutinising the rationale for billing amounts issued by these doctors and identifying treatments prescribed by these doctors over time.

About this document

This document sets out the analytical steps taken to generate the summary insights above, with findings in the left column and SQL code in the right-hand column.

Data validation and cleaning

Steps to validate data linkages included matching primary and foreign keys in the patient, billing and treatment tables, and matching appointment, treatment and billing dates. No orphan records were identified.

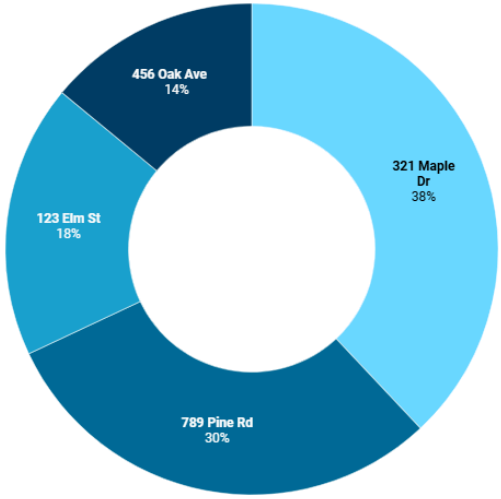
```sql
-- Check data linkages between the patient and billing tables
SELECT *
FROM data-analytics-bootcamp-4.Hospital_management.billing as b
WHERE b.patient_id NOT IN (SELECT p.patient_id FROM
data-analytics-bootcamp-4.Hospital_management.patients as p);

-- Check data linkages between the billing and treatment tables
SELECT *
FROM data-analytics-bootcamp-4.Hospital_management.billing as b
WHERE b.treatment_id NOT IN (SELECT t.treatment_id FROM
data-analytics-bootcamp-4.Hospital_management.treatments as t);
```

There was also no evidence of additional whitespace in the primary or foreign keys of the datasets.

```sql
-- Check for differences between appointment dates, treatment dates and billing
dates
SELECT a.appointment_date, DATE_DIFF(a.appointment_date, t.treatment_date, day) AS
days_after_appointment, t.treatment_date, b.bill_date, DATE_DIFF(t.treatment_date,
b.bill_date, day) AS days
FROM data-analytics-bootcamp-4.Hospital_management.treatments as t
JOIN data-analytics-bootcamp-4.Hospital_management.billing as b
  ON t.treatment_id = b.treatment_id
JOIN data-analytics-bootcamp-4.Hospital_management.appointments as a
  ON t.appointment_id = a.appointment_id
ORDER BY days_after_appointment DESC;
```

However, only four addresses are listed across the 50 patients, indicating that addresses have been recorded incorrectly:




```
-- Show the number of distinct addresses listed in the patients table
SELECT DISTINCT address, COUNT(address) AS number_of_addresses
FROM data-analytics-bootcamp-4.Hospital_management.patients as P
GROUP BY address
ORDER BY COUNT(address) DESC;
```

There are also data quality issues in the email field, with 11 email addresses duplicated across multiple IDs.

Number of patient IDs	Number of email addresses
50	39

```
-- Show number of patients and number of email addresses
SELECT COUNT(DISTINCT patient_id) AS number_of_patients, COUNT(DISTINCT email)
AS number_of_email_addresses
FROM data-analytics-bootcamp-4.Hospital_management.patients as p;
```

The same email address is listed for multiple patient IDs with the same name. For example:

Patient ID	Full name	Email	Number of occurrences of email
P036	Michael Wilson	michael.wilson@mail.com	3
P021	Michael Wilson	michael.wilson@mail.com	3
P033	Michael Wilson	michael.wilson@mail.com	3

```
-- Show patient name and ID for emails associated with michael.taylor@mail.com
SELECT p.patient_id,
      CONCAT(p.first_name, ' ', p.last_name) AS full_name,
      email,
       COUNT(email) OVER (PARTITION BY email) as number_emails
FROM data-analytics-bootcamp-4.Hospital_management.patients as p
ORDER BY number_emails DESC
LIMIT 3;
```

Ten patients are listed as having attended appointments more than 14 days before they were registered at the hospital. Even allowing for a two-week gap between their appointment and registration date, this suggests that there may be problems with data entry.

Patient ID	Full name	Appointments attended before registration date
P033	Michael Wilson	4
P029	David Smith	3
P012	Laura Davis	3
P035	David Wilson	3
P049	David Moore	3
P028	Alex Moore	3
P014	Alex Taylor	2
P034	Alex Smith	2
P048	Emily Miller	2
P050	Laura Wilson	1

```
-- Show patients who have attended appointments before their registration date
SELECT p.patient_id,
      CONCAT(p.first_name, ' ', p.last_name) AS full_name,
      COUNT(p.patient_id) AS appointments_before_registration
FROM data-analytics-bootcamp-4.Hospital_management.appointments as A
JOIN data-analytics-bootcamp-4.Hospital_management.patients as P
  ON a.patient_id = p.patient_id
WHERE (DATE_DIFF(p.registration_date, a.appointment_date, day)) > 14
GROUP BY p.patient_id, full_name
ORDER BY COUNT(p.patient_id) DESC;
```

Finally, for 32% of the patients (16 of the 50), either those with names commonly associated with the female gender are assigned the gender ‘M’, or those with names commonly associated with the male gender are assigned the gender ‘F’.

Please note that this does *not* necessarily indicate that the patients’ gender has been recorded incorrectly. However, in the context of other data quality issues, this should be considered alongside the other points outlined above.

```
-- Count common ‘female’ names assigned to the male gender and common ‘male’
names assigned to the female gender
SELECT COUNT(DISTINCT p.patient_id) AS names_for_checking
FROM data-analytics-bootcamp-4.Hospital_management.patients as P
WHERE (p.gender = 'M' AND p.first_name IN('Sarah','Laura','Linda')) OR
(p.gender = 'F' AND p.first_name IN ('David','Michael','Robert'));
```

High-risk patients

Seven patients are particularly high-risk from an insurance perspective because they have attended a significant number of appointments at a cost-per-appointment of more than $1,700. (The average cost per appointment across the hospital is $550.)

Patient ID	Full name	Total billed ($)	Number of appointments	Amount billed per appointment ($)	Difference vs hospital-wide average amount billed per appointment ($)
P012	Laura Davis	30,053.08	10	3,005.31	2,455.16
P049	David Moore	23,554.06	7	3,364.87	2,814.72
P016	Michael Taylor	22,967.94	7	3,281.13	2,730.98
P036	Michael Wilson	21,583.56	7	3,083.37	2,533.22
P025	Robert Wilson	19,513.17	5	3,902.63	3,352.48
P005	David Wilson	18,609.91	8	2,326.24	1,776.09
P035	David Wilson	18,407.42	7	2,629.63	2,079.48

They have also highest billed amounts to the hospital overall, totalling more than $150,000.

Total billed by high-risk patients above ($)
154,689.14

```sql
- Show patients that attended the highest number of appointments and the amount
billed for each of those appointments
WITH stats AS (
  SELECT  ROUND((SUM(DISTINCT b.amount))/COUNT(a.appointment_id),2) as
AVG_COST_PER_APPOINTMENT
FROM data-analytics-bootcamp-4.Hospital_management.billing as b
JOIN data-analytics-bootcamp-4.Hospital_management.patients as p
    ON b.patient_id = p.patient_id
JOIN data-analytics-bootcamp-4.Hospital_management.appointments as a
    ON a.patient_id = p.patient_id
)
SELECT b.patient_id,
      CONCAT(p.first_name, ' ', p.last_name) AS full_name,
      ROUND(SUM(DISTINCT b.amount),2) AS total_billed,
      COUNT(DISTINCT a.appointment_id) AS number_appointments,
      ROUND(SUM(b.amount)/COUNT(a.appointment_id),2) AS cost_per_appointment,
ROUND(SUM(b.amount)/COUNT(a.appointment_id)-s.AVG_COST_PER_APPOINTMENT,2) AS
difference_from_average_appointment_cost
FROM data-analytics-bootcamp-4.Hospital_management.billing as b
JOIN data-analytics-bootcamp-4.Hospital_management.patients as p
    ON b.patient_id = p.patient_id
JOIN data-analytics-bootcamp-4.Hospital_management.appointments as a
    ON a.patient_id = p.patient_id
JOIN data-analytics-bootcamp-4.Hospital_management.treatments as t
  ON a.appointment_id = t.appointment_id
JOIN stats as s ON b.patient_id = p.patient_id
GROUP BY b.patient_id, full_name, s.AVG_COST_PER_APPOINTMENT
ORDER BY total_billed DESC;

-- Show the total billed by the identified high-risk patients
SELECT ROUND(SUM(b.amount),2) AS total_billed
FROM data-analytics-bootcamp-4.Hospital_management.billing as b
JOIN data-analytics-bootcamp-4.Hospital_management.patients as p
    ON b.patient_id = p.patient_id
WHERE  p.patient_id IN ('P012', 'P049', 'P016', 'P036', 'P025', 'P005', 'P035');
```

No one type of appointment seems to be generating significantly higher billed amounts.

Reason for visit	Number of appointments	Total amount billed ($)
Checkup	45	621328.9
Consultation	43	585458.36
Therapy	42	600879.41
Follow-up	41	549876.74
Emergency	29	390746.75

```sql
-- Show amount billed for each type of reason for visit
SELECT
  a.reason_for_visit,
  COUNT(DISTINCT a.appointment_id) AS number_appointments,
  ROUND(SUM(b.amount),2) AS total_billed,
FROM data-analytics-bootcamp-4.Hospital_management.billing AS b
JOIN data-analytics-bootcamp-4.Hospital_management.patients AS p
  ON b.patient_id = p.patient_id
JOIN data-analytics-bootcamp-4.Hospital_management.appointments AS a
  ON a.patient_id = p.patient_id
JOIN data-analytics-bootcamp-4.Hospital_management.treatments AS t
  ON t.treatment_id = b.treatment_id
GROUP BY a.reason_for_visit
ORDER BY number_appointments DESC;
```

Multiple patients have received more than three advanced protocol treatments, making them higher-risk from a medical standpoint.

Patient ID	Full name	Advanced protocol treatments
P019	Sarah Miller	4
P001	David Williams	3
P035	David Wilson	3
P023	Linda Johnson	3
P007	Alex Johnson	3

```sql
-- Show top five patients who have received 3 or more advanced protocol
treatments
SELECT a.patient_id,
        CONCAT(p.first_name, ' ', p.last_name) AS full_name, t.description,
        COUNT(t.description) as number_of_treatments
FROM data-analytics-bootcamp-4.Hospital_management.treatments as t
JOIN data-analytics-bootcamp-4.Hospital_management.appointments as a
    ON t.appointment_id = a.appointment_id
JOIN data-analytics-bootcamp-4.Hospital_management.patients as p
    ON a.patient_id = p.patient_id
WHERE t.description = 'Advanced protocol'
GROUP BY a.patient_id, full_name, t.description
HAVING COUNT(t.description) > 2
ORDER BY COUNT(t.description) DESC
LIMIT 5;
```

However, the seven high-risk patients mainly received basic- or standard-level treatments, not advanced ones.

Given the seven high-risk patients' high cost per appointment, they might be expected to have mainly undertaken Advanced Protocol treatments. However, more than 70% of the treatments given to six of the seven patients were either Basic Screenings or Standard Procedures.

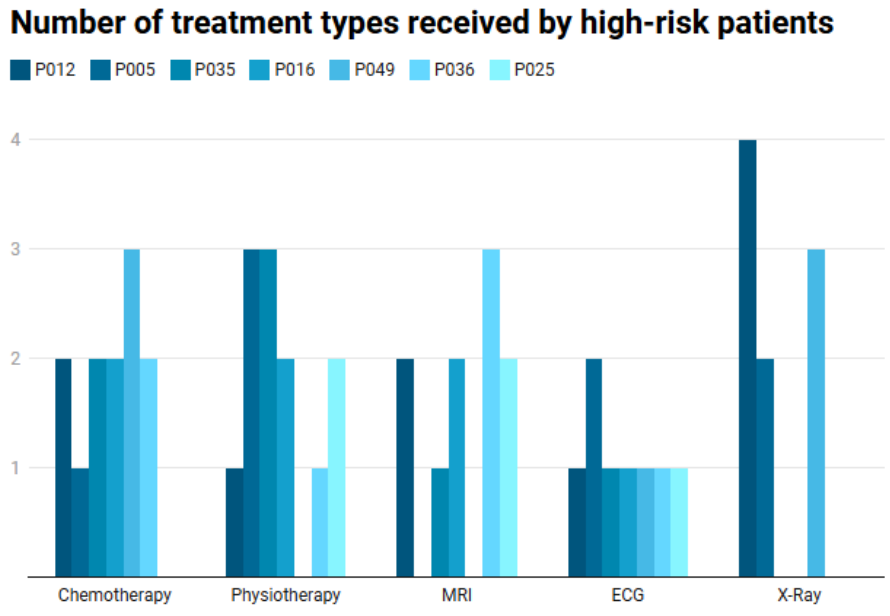
Patient ID	Full name	Number of treatments	Number Advanced Protocol treatments	Number of Standard Procedure treatments	Number of Basic Screenings	Percentage treatments that were standard or basic-level (%)
P012	Laura Davis	10	0	6	4	100
P049	David Moore	7	1	5	1	85.7
P036	Michael Wilson	7	1	4	2	85.7
P025	Robert Wilson	5	1	2	2	80
P016	Michael Taylor	7	2	2	3	71.4
P005	David Wilson	8	2	4	2	75
P035	David Wilson	7	3	0	4	57.1

This suggests that the treatments given to the patients were abnormally high in cost given their level of complexity.

```sql
-- Show level of treatment given to selected high-risk patients
SELECT p.patient_id, CONCAT(p.first_name, ' ', p.last_name) AS full_name,
    COUNT(t.description) AS number_treatments,
    SUM(CASE
        WHEN description = 'Advanced protocol' THEN 1
        ELSE 0
        END) AS number_advanced_protocol,
    SUM(CASE
        WHEN description = 'Standard procedure' THEN 1
        ELSE 0
        END) AS number_standard_procedure,
    SUM(CASE
        WHEN description = 'Basic screening' THEN 1
        ELSE 0
        END) AS number_basic_screening,
    ROUND(SUM(CASE
        WHEN description = 'Basic screening' THEN 1
        WHEN description = 'Standard procedure' THEN 1
        ELSE 0
        END)/COUNT(t.description)*100,1) AS percentage_basic_or_standard
FROM data-analytics-bootcamp-4.Hospital_management.treatments as t
JOIN data-analytics-bootcamp-4.Hospital_management.appointments as a
    ON t.appointment_id = a.appointment_id
JOIN data-analytics-bootcamp-4.Hospital_management.patients as p
    ON a.patient_id = p.patient_id
WHERE p.patient_id IN ('P012', 'P049', 'P016', 'P036', 'P025', 'P005', 'P035')
GROUP BY p.patient_id, full_name
ORDER BY number_advanced_protocol;
```

We now need to identify potential reasons for this high cost-per-appointment.

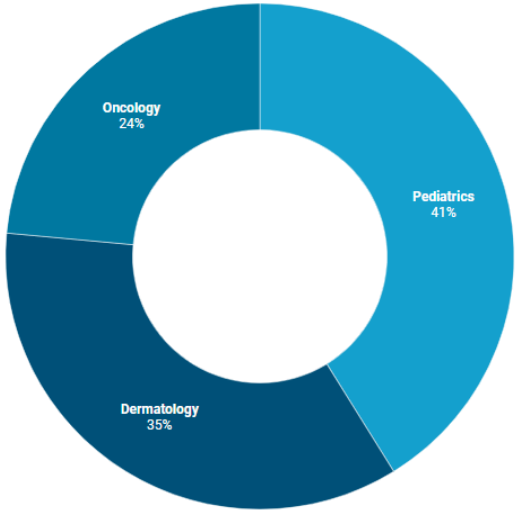
First, no treatment type was particularly strongly associated with high-cost appointments:



This suggests that no one treatment type was responsible for all abnormally high billing amounts.

```sql
-- Show type of treatment given to high-risk patients
SELECT p.patient_id, CONCAT(p.first_name, ' ', p.last_name) AS full_name,
    COUNT(t.description) AS number_treatments,
    SUM(CASE
        WHEN t.treatment_type = 'Physiotherapy' THEN 1
        ELSE 0
        END) AS number_physiotherapy,
    SUM(CASE
        WHEN t.treatment_type = 'X-Ray' THEN 1
        ELSE 0
        END) AS number_Xray,
    SUM(CASE
        WHEN t.treatment_type = 'ECG' THEN 1
        ELSE 0
        END) AS number_ECG,
    SUM(CASE
        WHEN t.treatment_type = 'Chemotherapy' THEN 1
        ELSE 0
        END) AS number_Chemotherapy,
    SUM(CASE
        WHEN t.treatment_type = 'MRI' THEN 1
        ELSE 0
        END) AS number_MRI
FROM data-analytics-bootcamp-4.Hospital_management.treatments as t
JOIN data-analytics-bootcamp-4.Hospital_management.appointments as a
    ON t.appointment_id = a.appointment_id
JOIN data-analytics-bootcamp-4.Hospital_management.patients as p
    ON a.patient_id = p.patient_id
WHERE p.patient_id IN ('P012', 'P049', 'P016', 'P036', 'P025', 'P005', 'P035')
GROUP BY p.patient_id, full_name
ORDER BY number_treatments DESC;
```

The high-risk patients had the most appointments with doctors specialising in pediatrics (19 appointments), followed by dermatology (16) and oncology (11).




```
-- Show specialization of the doctors that saw the highest numbers of selected high-risk patients
SELECT d.specialization,
    COUNT(DISTINCT a.appointment_id) AS total_appointments,
    ROUND(SUM(b.amount),2) AS total_billed
FROM data-analytics-bootcamp-4.Hospital_management.doctors as d
JOIN data-analytics-bootcamp-4.Hospital_management.appointments as a
    ON d.doctor_id = a.doctor_id
JOIN data-analytics-bootcamp-4.Hospital_management.patients as p
    ON a.patient_id = p.patient_id
JOIN data-analytics-bootcamp-4.Hospital_management.billing as b
    ON b.patient_id = p.patient_id
WHERE a.patient_id IN ('P012', 'P049', 'P016', 'P036', 'P025', 'P005', 'P035')
GROUP BY d.specialization
ORDER BY total_appointments DESC;
```

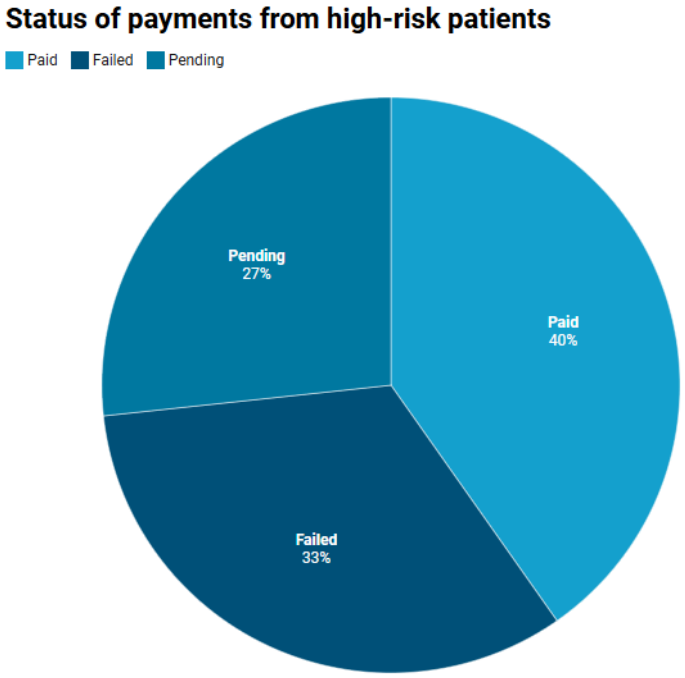
However, none of the high-risk patients are aged below 16 (see table). It is therefore unclear why they were seeing specialists in children's health.

Patient ID	Full name	Age band
P016	Michael Taylor	18-25
P036	Michael Wilson	26-34
P035	David Wilson	26-34
P012	Laura Davis	26-34
P049	David Moore	35-05
P025	Robert Wilson	51-65
P005	David Wilson	51-65

```
-- Calculate age of high-risk patients
SELECT patient_id,
        date_of_birth,
        CONCAT(p.first_name, ' ', p.last_name) AS full_name,
        FLOOR(DATE_DIFF('2026-02-10',date_of_birth, DAY)/365) AS age
FROM data-analytics-bootcamp-4.Hospital_management.patients as P
WHERE patient_id IN ('P012', 'P049', 'P016', 'P036', 'P025', 'P005', 'P035')
ORDER BY age;
```

The seven patients have also failed to pay for more than $80,000 in treatment costs – more than half of the total amount billed to them.

The seven patients are collectively responsible for $44,662 in failed billings and $36,092 in pending billings - a total of $80,755.



Some of these failed payments are particularly high. Three of these eight patients have payments of over $8,800 that have failed.

Patient ID	Full name	Total failed billings ($)
P012	Laura Davis	10,404.07
P005	David Wilson	10,266.94
P036	Michael Wilson	8,884.25

```
-- Show failed and pending billings linked to high-risk patients
SELECT ROUND(SUM(b.amount),2) AS total_failed_pending_billings
FROM data-analytics-bootcamp-4.Hospital_management.billing as B
JOIN data-analytics-bootcamp-4.Hospital_management.patients as P
    ON b.patient_id = p.patient_id
WHERE b.payment_status = 'Failed' AND p.patient_id IN ('P012', 'P049', 'P016',
'P036', 'P025', 'P005', 'P035');

SELECT ROUND(SUM(b.amount),2) AS total_pending_billings
FROM data-analytics-bootcamp-4.Hospital_management.billing as B
JOIN data-analytics-bootcamp-4.Hospital_management.patients as P
    ON b.patient_id = p.patient_id
WHERE b.payment_status IN ('Pending', 'Failed') AND p.patient_id IN ('P012',
'P049', 'P016', 'P036', 'P005', 'P035');
```

```
-- Show highest failed payments by high-risk patients
SELECT b.patient_id,
        CONCAT(p.first_name, ' ', p.last_name) AS full_name,
        ROUND(SUM(b.amount),2) AS total_failed_billings
FROM data-analytics-bootcamp-4.Hospital_management.billing as B
JOIN data-analytics-bootcamp-4.Hospital_management.patients as P
    ON b.patient_id = p.patient_id
WHERE b.payment_status = 'Failed' AND p.patient_id IN ('P012', 'P049', 'P016',
'P036', 'P005', 'P035')
GROUP BY b.patient_id, full_name
GROUP total_failed_billings > 7000
ORDER BY total_failed_billings DESC;
```

Multiple insurance bills have also been issued that are failed or pending in high amounts, several to one of the high-risk patients identified above (Michael Wilson, patient ID PO36).

Bill ID	Patient ID	Amount billed ($)	Payment status
B130	P017	4966.18	Failed
B173	P047	4890.25	Pending
B004	P025	4799.86	Failed
B042	P036	4781.32	Pending
B147	P014	4716.31	Failed
B012	P029	771.2	Pending
B113	P036	770.64	Pending

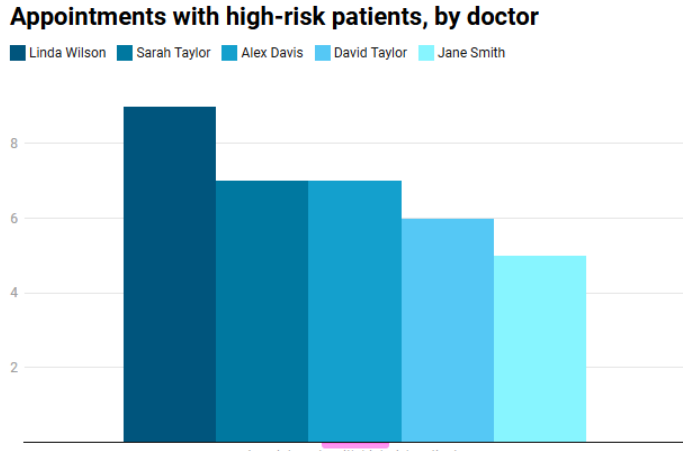
```sql
-- Show insurance payments that are unusually high and are either failed or pending
WITH stats AS (
    SELECT
        AVG(amount) AS mean,
        STDDEV(amount) AS stddev
    FROM data-analytics-bootcamp-4.Hospital_management.billing AS b
)
SELECT bill_id, patient_id, amount, payment_status
FROM data-analytics-bootcamp-4.Hospital_management.billing, stats
WHERE ABS(amount - mean) > 1.5 * STDDEV AND payment_status IN
('Failed','Pending') AND payment_method = 'Insurance'
ORDER BY amount DESC;
```

MedCare Plus is being billed for more than double the other insurance providers at the hospital ($1.4 million).

Insurance provider	Number of appointments	Amount billed ($)
MedCare Plus	84	1,375,998.36
WellnessCorp	58	627,555.7
PulseSecure	36	456,379.01
HealthIndia	22	288,357.09

```sql
-- Categorise insurance providers by number of appointments and total amount billed
SELECT
  p.insurance_provider,
  COUNT(DISTINCT a.appointment_id) AS number_appointments,
  ROUND(SUM(b.amount),2) AS total_billed,
FROM data-analytics-bootcamp-4.Hospital_management.billing AS b
JOIN data-analytics-bootcamp-4.Hospital_management.patients AS p
  ON b.patient_id = p.patient_id
JOIN data-analytics-bootcamp-4.Hospital_management.appointments AS a
  ON a.patient_id = p.patient_id
JOIN data-analytics-bootcamp-4.Hospital_management.treatments AS t
  ON t.treatment_id = b.treatment_id
GROUP BY p.insurance_provider
ORDER BY total_billed DESC;
```

Two doctors (Linda Wilson, Oncology) and (Sarah Taylor, Dermatology) were particularly likely to see the seven high-risk patients.




```sql
-- Show doctors that saw the selected high-risk patients most frequently
SELECT d.doctor_id, CONCAT(d.first_name, ' ', d.last_name) AS full_name,
COUNT(a.appointment_id) AS total_appointments
FROM data-analytics-bootcamp-4.Hospital_management.doctors as d
JOIN data-analytics-bootcamp-4.Hospital_management.appointments as a
    ON d.doctor_id = a.doctor_id
JOIN data-analytics-bootcamp-4.Hospital_management.patients as p
    ON a.patient_id = p.patient_id
WHERE p.patient_id IN ('P012', 'P049', 'P016', 'P036', 'P025', 'P005', 'P035')
GROUP BY d.doctor_id, full_name
ORDER BY total_appointments DESC
LIMIT 5;
```

These two doctors were among those who billed the most to MedCare Plus.

Doctor ID	Full name	Total billed per doctor ($)
D005	Sarah Taylor	203,907.52
D006	Alex Davis	192,174.85
D001	David Taylor	159,317.95
D002	Jane Davis	140,068.88
D007	Robert Davis	132,894.95
D010	Linda Wilson	128,925.09
D004	David Jones	110,900.00
D003	Jane Smith	110,802.13

```sql
-- Identify doctors who billed most to MedCare Plus
SELECT d.doctor_id,
        CONCAT(d.first_name,' ',d.last_name) as full_name,
        ROUND(SUM(b.amount),2) as total_billed_per_doctor
FROM data-analytics-bootcamp-4.Hospital_management.doctors as d
JOIN data-analytics-bootcamp-4.Hospital_management.appointments as a
    ON d.doctor_id = a.doctor_id
JOIN data-analytics-bootcamp-4.Hospital_management.patients as p
    ON a.patient_id = p.patient_id
JOIN data-analytics-bootcamp-4.Hospital_management.billing as b
    ON b.patient_id = p.patient_id
WHERE p.insurance_provider = 'MedCare Plus'
GROUP BY d.doctor_id, full_name
ORDER BY total_billed_per_doctor DESC;
```

Potential next steps

To conduct further investigations into overspending and potential fraudulent activity, actions could include:

- Analysing the potential for overspending in basic- and standard-level treatments. This could involve establishing an agreed cost range and conducting additional checks on any billing amounts that fall outside that range.
- Assessing the range and spread of billing amounts to identify outliers and anomalies, and treatment types, doctors and specialisations that may be associated with them.
- Analyse billing patterns on a month-by-month basis to identify spikes and/or variations, particularly those attributable to high-risk patients.